

# Food Insecurity Forecast - Model Comparison

Two models compared: the original model and the enhanced model. Both are population-weighted ridge regression transition models trained on national ZCTA data from 2020-2024 and used to forecast Feeding Tampa Bay ZIP/ZCTA-level food insecurity rates for 2025-2027.

**Short answer.** The enhanced model is the better candidate based on the available validation test: its Feeding Tampa Bay average error is about **44% lower** than the original model. Both models predicted the 2024 holdout year too high on average, but the enhanced model was closer.

## Summary SIDE-BY-SIDE

| MODEL          | MODEL TYPE                                             | WHAT IT USES                                                                                                                 | VALIDATION RESULT                                              | READ THIS AS                                                 |
|----------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------|
| Original model | Population-weighted ridge regression transition model. | Prior-year food insecurity, unemployment, poverty, income, demographics, homeownership, disability, and a linear year trend. | FTB holdout MAE 0.0219; predicted 17.4% when actual was 15.4%. | Simple and transparent, but too high in the 2024 local test. |
| Enhanced model | Population-weighted ridge regression transition model. | Everything in the original model except the year trend, plus county food cost, county context, and ALICE hardship measures.  | FTB holdout MAE 0.0123; predicted 16.2% when actual was 15.4%. | More accurate in the 2024 test, though still slightly high.  |

Lower MAE / RMSE is better. These errors are measured on a 0-1 rate scale, so 0.0123 means about 1.2 percentage points of average error.

# How the Models Work

$$\text{RATE}[T] \leftarrow \text{RATE}[T-1] + \text{DRIVERS}$$

Both models use a transition-model structure. In plain English, each model starts with what food insecurity looked like last year, then adjusts the estimate using economic, demographic, and local-condition variables. For the forecast, the models are recursive: 2025 uses the observed 2024 rate, 2026 uses the predicted 2025 rate, and 2027 uses the predicted 2026 rate.

**How to read the years.** The models train on observed years **2020-2024**. The validation test hides 2024, trains on earlier transitions, then asks: how close would the model have come to the 2024 values we now know?

# Projecting Future Inputs

DRIVERS FOR 2025-2027

Before either model can forecast food insecurity, it needs future values for the input drivers, such as unemployment, poverty, income, homeownership, disability, population, demographics, and, for the enhanced model, county food-cost and ALICE variables. Because independent future forecasts for every ZIP/ZCTA-county row are not included in the source data, the models create internal driver projections.

| STEP                                               | WHAT HAPPENS                                                                                                                            |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Start with the latest observed row                 | The 2024 value is used as the baseline for each Feeding Tampa Bay ZIP/ZCTA-county row.                                                  |
| Estimate a recent trend                            | For each row and each driver, the model estimates a simple straight-line trend from the observed 2020-2024 history.                     |
| Extend that trend forward                          | The trend is carried forward into 2025-2027. For example, a driver that has been rising is projected to keep rising unless constrained. |
| Keep values in valid ranges                        | Rate-like inputs are kept between 0% and 95%, and projected population is not allowed to go below zero.                                 |
| Use projected drivers in the food insecurity model | Each forecast year uses that year's projected drivers plus the prior-year food insecurity rate.                                         |

**Original vs. enhanced input projections.** The original model uses the simple linear driver projection with basic validity checks. The enhanced model uses the same linear projection, then adds stabilizing guardrails: key annual driver changes are capped, population is constrained to a broad +/-3% annual band around the 2024 baseline, and structural fields such as SNAP threshold, rural/urban code, and ALICE data availability are held constant unless new data is supplied.

## Performance

FEEDING TAMPA BAY HOLDOUT

| QUESTION                     | ORIGINAL MODEL        | ENHANCED MODEL        | WINNER              |
|------------------------------|-----------------------|-----------------------|---------------------|
| Average local miss           | 2.2 percentage points | 1.2 percentage points | Enhanced            |
| Typical large-error penalty  | RMSE 0.0244           | RMSE 0.0153           | Enhanced            |
| Bias direction               | Predicted too high    | Predicted too high    | Enhanced is closer  |
| Actual weighted 2024 rate    | 15.4%                 | 15.4%                 | Same test           |
| Predicted weighted 2024 rate | 17.4%                 | 16.2%                 | Closer:<br>enhanced |

The enhanced model also improves national 2024 holdout MAE from 0.0229 to 0.0151 and improves in-sample weighted MAE from 0.0122 to 0.0100.

## What Changed in the Enhanced Model ADDED DRIVERS

The enhanced model adds variables meant to describe local cost pressure and household financial strain, not just ZIP-level demographics. This gives the model more context, but it also makes the model harder to explain and increases the chance of overlapping signals.

| GROUP                   | ORIGINAL MODEL                               | ADDED IN ENHANCED MODEL                                                         | WHY IT MATTERS                                        |
|-------------------------|----------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|
| Food insecurity history | Prior-year food insecurity rate              | No change                                                                       | Recent local history is usually the strongest signal. |
| Economic conditions     | Unemployment, poverty, median income         | No change                                                                       | Captures financial stress and resources.              |
| Demographics            | Black share, Hispanic share, disability rate | County child share; county White non-Hispanic share                             | Adds broader county composition context.              |
| Housing                 | Homeownership rate                           | No change                                                                       | Often proxies for wealth and residential stability.   |
| Food costs              | Not included                                 | Cost per meal; food cost index                                                  | Captures whether food is more expensive locally.      |
| County context          | Not included                                 | SNAP threshold; rural/urban code                                                | Adds policy and geography context.                    |
| ALICE hardship          | Not included                                 | Financial insecurity, poverty households, survival-budget thresholds, data flag | Adds household budget pressure measures.              |

## Forecast Pattern 2025-2027

| YEAR | ORIGINAL AVG RATE | ENHANCED AVG RATE | ENHANCED MIN-MAX | DIFFERENCE |
|------|-------------------|-------------------|------------------|------------|
| 2025 | 17.5%             | 17.0%             | 0.1% - 64.3%     | -0.4 pts   |
| 2026 | 19.3%             | 19.2%             | 0.8% - 67.3%     | -0.1 pts   |
| 2027 | 21.0%             | 21.7%             | 2.1% - 70.2%     | +0.7 pts   |

Both outputs contain 259 Feeding Tampa Bay ZCTA/county rows per year. The enhanced model starts slightly lower than the original model in 2025, stays slightly lower in 2026, then ends higher by 2027.

# Recommendation

MODEL USE

**Recommended candidate: enhanced model.** It is the better-performing model in the available holdout test, and it includes important local context around food costs and financial hardship. Use the original model as a benchmark because it is simpler and easier to explain.

| NEXT STEP                              | WHY                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Review or calibrate bias               | The enhanced model predicted 2024 too high by about 0.8 percentage points on average for Feeding Tampa Bay rows.      |
| Test a simpler enhanced model          | Some enhanced-model variables overlap conceptually; a smaller version may be nearly as accurate and easier to defend. |
| Keep local validation separate         | National performance is useful, but the decision audience cares most about the Feeding Tampa Bay region.              |
| Show uncertainty in planning materials | Forecasts are directional estimates, not exact future counts.                                                         |

# Plain-English Glossary

TERMS USED ABOVE

| TERM             | MEANING                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------|
| ZCTA             | Census approximation of a ZIP Code area.                                                              |
| Holdout test     | A test where the model predicts a year we already know, so we can measure its error.                  |
| MAE              | Mean absolute error: the average size of the prediction miss. Lower is better.                        |
| RMSE             | A second error measure that penalizes bigger misses more heavily. Lower is better.                    |
| Bias             | Whether predictions tend to be too high or too low.                                                   |
| Ridge regression | A linear model with a stabilizing penalty so correlated variables do not swing the forecast too much. |

## Technical Appendix COEFFICIENTS

The models standardize predictors before fitting ridge regression. That means each coefficient reflects a one-standard-deviation change in the input, not a one-unit raw change. Treat these as predictive associations, not causal effects.

| ORIGINAL MODEL DRIVER           | COEF.   | WHAT IT MEANS                                |
|---------------------------------|---------|----------------------------------------------|
| Prior-year food insecurity rate | +0.0298 | A ZIP/ZCTA's recent food insecurity history. |
| Unemployment rate               | +0.0055 | Local unemployment.                          |
| Poverty rate                    | +0.0125 | Share of residents below poverty.            |
| Percent Black                   | -0.0039 | ZIP/ZCTA demographic composition.            |
| Percent Hispanic                | +0.0025 | ZIP/ZCTA demographic composition.            |
| Median income, log              | -0.0021 | Income level, log-scaled.                    |
| Homeownership rate              | -0.0045 | Share of homes owner-occupied.               |
| Disability rate                 | +0.0073 | Share of residents with a disability.        |
| Centered year                   | +0.0085 | Linear time trend in the original model.     |

| ENHANCED MODEL DRIVER           | COEF.   | WHAT IT MEANS                                |
|---------------------------------|---------|----------------------------------------------|
| Prior-year food insecurity rate | +0.0452 | A ZIP/ZCTA's recent food insecurity history. |
| Unemployment rate               | +0.0027 | Local unemployment.                          |
| Poverty rate                    | +0.0064 | Share of residents below poverty.            |
| Percent Black                   | -0.0018 | ZIP/ZCTA demographic composition.            |
| Percent Hispanic                | +0.0037 | ZIP/ZCTA demographic composition.            |
| Median income, log              | +0.0036 | Income level, log-scaled.                    |
| Homeownership rate              | -0.0038 | Share of homes owner-occupied.               |
| Disability rate                 | +0.0054 | Share of residents with a disability.        |
| County child population share   | -0.0002 | Share of county residents who are children.  |
| County % white, non-Hispanic    | +0.0015 | County demographic composition.              |
| County cost per meal            | +0.0218 | Estimated average meal cost.                 |

| ENHANCED MODEL DRIVER           | COEF.   | WHAT IT MEANS                                      |
|---------------------------------|---------|----------------------------------------------------|
| County food cost index          | -0.0213 | County food cost relative to national average.     |
| County SNAP threshold           | -0.0019 | SNAP eligibility income threshold.                 |
| Rural-urban code                | -0.0008 | County rural/urban classification.                 |
| ALICE financial insecurity rate | -0.0001 | Household hardship measure from ALICE data.        |
| ALICE poverty household rate    | +0.0003 | Household poverty measure from ALICE data.         |
| ALICE threshold under 65        | +0.0009 | Survival-budget threshold for under-65 households. |
| ALICE threshold 65+             | +0.0001 | Survival-budget threshold for 65+ households.      |
| Has ALICE data                  | -0.0007 | Flag for whether county ALICE data was available.  |

Forecast output sources were filtered to 2025-2027 for this report.